

Manh Nguyen

(+61) 414 343 129 | nducmanh08@gmail.com | Sydney, NSW
[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Software Engineer with hands-on experience developing full-stack applications and AI-powered solutions using **Spring Boot, TypeScript, Python**, and **cloud technologies**. Skilled in developing **API architecture, database systems**, and **CI/CD automation**, with a focus on building **user-centric, scalable software products**. Passionate about leveraging real-world problems, contributing to innovative teams, while continuously learning.

EXPERIENCE

AI Full Stack Developer

Jul. 2024 – Oct. 2025

Home Invease

- Developed **AI-powered household inventory management application** enabling users to organize items through **object recognition** and **receipt scanning**, using **React Native** for cross-platform mobile interface.
- Designed **Spring Boot microservices** with **RESTful APIs** to orchestrate inventory management, AI inference, and secure authentication using **OAuth2** across mobile and web applications.
- Architected **RAG pipeline** combining **ChromaDB** vector search with **Ollama LLMs**, enhancing context-aware inventory search and **reducing query latency by 30%**.
- Fine-tuned **image classification model** to recognize household items from photos and paired it with **Tesseract OCR** to extract receipt data, populating inventory records and eliminating manual entry.
- Collaborated across the **Agile** team from system design, **JUnit** testing, deployment, and iterative feature delivery using **Git-based workflows** and **Microsoft Azure cloud services**.

PROJECTS

Link AI Assistant | *React, Next.js, Express.js, TypeScript, Claude, OAuth2*

[GitHub](#)

- Architected an **AI-powered file management assistant** integrating **Google Drive API** to automate file organization and storage optimization.
- Designed **weighted scoring engine** combining metadata features to calculate **confidence scores** and **relationship** between files.
- Developed an integrated **Claude Code pipeline** that ranks file relationships and provides organization suggestions based on **similarity confidence scores**.
- Established **CI/CD pipelines** and deployed as website and browser extension using **Chrome Extension Manifest V3** and **GitHub Actions**.

Loan Default Prediction Model | *Python, PostgreSQL, Apache Spark, Machine Learning*

[GitHub](#)

- Engineered an **end-to-end ML decision-support platform** using **Apache Spark** trained on over **850,000 customer records**.
- Integrated **data preprocessing, predictive modelling**, and **RESTful API services** for automated credit risk assessment.
- Built a **feature engineering pipeline** generating financial risk indicators to improve prediction reliability.

EDUCATION

University of Wollongong

Wollongong, NSW

Bachelor of Computer Science (Artificial Intelligence and Big Data)

Jul. 2022 – Jun. 2025

- Distinction Grade:** Advanced Programming, Database Systems, Modern Artificial Intelligence, AI & Machine Learning, Big Data Management, Big Data Mining Technologies.

TECHNICAL SKILLS

Languages: Java, Python, C++, C#, TypeScript, JavaScript, SQL, R

Software Engineering: Spring Boot, React, Next.js, Node.js, RESTful APIs, PostgreSQL, MongoDB, Firebase, ChromaDB, OAuth2, JWT

AI & Big Data: LLMs, RAG, NLP, Computer Vision, TensorFlow, PyTorch, Ollama, Apache Spark, Hadoop

Cloud & DevOps: Microsoft Azure, Google Cloud Platform, Docker, Git, GitHub Actions